PROJECT PROPOSAL

CampusConnect: Campus Event & Club Management Portal

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Registration Number: [Student Registration Number]

Department: Department of Computer Science & Engineering

Faculty Guide: [Faculty Guide Name, Designation]

Subjects Involved: Full Stack Development, Backend Development, Advanced Database Systems

Project Category: Full-Stack Web Application / Academic Microproject

1. Problem Statement

University campuses feature active student societies organizing technical hackathons, cultural festivals, and seminars. However, registration management is fragmented across unlinked spreadsheets and messaging channels. This leads to seat overbooking, lack of duplicate checking, and frustration for both students and event coordinators.

2. Proposed Solution

CampusConnect introduces an integrated web portal where students can explore student societies, browse events with live capacity tracking, and secure verified seat reservations. Organizers receive an administrative console to publish events, configure capacity limits, and inspect attendee rosters in real-time.

3. Objectives & Scope

• Build a modern responsive SPA using React 18 and Vite. • Implement a layered Express backend with request correlation tracing and centralized error handling. • Enforce strict concurrency control using atomic compare-and-swap database transactions. • Design a normalized relational PostgreSQL schema adhering to 3NF/BCNF. • Deploy an append-only MongoDB document extension for event activity logging and analytical aggregation.

4. Technology Stack

• Frontend: React 18, Vite, React Router, Axios, TanStack Query, Custom CSS tokens. • Backend: Node.js, Express, Prisma ORM, Mongoose. • Primary Database: PostgreSQL (Relational) / SQLite (Dev runtime). • Extended Database: MongoDB (Document store for activity logs). • Testing: Node.js native test runner (`node --test`), Puppeteer browser automation.

5. Expected Deliverables

1. Complete, functional source code for frontend and backend. 2. Automated integration test suite validating health, CRUD, and concurrency limits. 3. Complete SQL schema DDL, normalization documentation, CTE queries, and window function demonstrations. 4. Three comprehensive academic project reports (Full Stack, Backend, ADBMS) in DOCX and PDF. 5. Project Proposal in DOCX and PDF. 6. Real browser screenshots demonstrating live workflows. 7. Final submission ZIP package.